WORKED SOLUTION

STEP I 2004, Question 14

Probability and Statistics • Counting Arrangements

STEP I 2004, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q14

The key idea

Picture the whole chest being emptied one coin at a time into a single row. The pirates simply cut this row at the two lead coins: the first pirate gets the gold before the first lead coin, the second the gold between the lead coins, and the third the gold after the second lead coin. Every ordering of the coins is equally likely, so everything comes down to counting where the two lead coins can sit.

Solution

Imagine all $n+2$ coins drawn out in a random order, even though the third pirate actually just scoops up what is left. That doesn't change who gets which coins. The draws are random, so each of the $\binom{n+2}{2}$ choices of the two positions (out of $n+2$) for the lead coins is equally likely.

Let A, B and C be the numbers of gold coins taken by the first, second and third pirates. Then $A+B+C=n$, and the row of coins looks like this:



Here gold coins are amber and lead coins are plum. The lead positions determine (A,B,C), and every triple of non-negative integers with $A+B+C=n$ comes from exactly one choice of positions. So the $\binom{n+2}{2}=\frac{1}{2}(n+1)(n+2)$ triples (A,B,C) are all equally likely.

Part (i)

The first pirate gets no gold exactly when $A=0$, that is, when the very first coin drawn is lead. That has probability $\frac{2}{n+2}$, since each of the $n+2$ coins is equally likely to come

out first. Hence

$$\mathrm{P}(A \geq 1) = 1 - \frac{2}{n+2} = \frac{n}{n+2}.$$

(Counting gives the same answer: $A = 0$ leaves $B + C = n$, which has $n + 1$ solutions, and $(n+1)/\frac{1}{2}(n+1)(n+2) = \frac{2}{n+2}$.)

Part (ii)

Ask yourself The second pirate starts drawing at a random time. Is she really no worse off than the first?

Yes, she is no worse off. We count the triples with $B = 0$: then $A + C = n$, which has $n + 1$ solutions, exactly as in part (i). So

$$\mathrm{P}(B \geq 1) = 1 - \frac{n+1}{\frac{1}{2}(n+1)(n+2)} = \frac{n}{n+2}.$$

In fact, since all triples (A, B, C) are equally likely, the three shares are *exchangeable*: every pirate has the same chance $\frac{n}{n+2}$ of getting some gold, even the third.

Part (iii)

We need A, B and C all to be at least 1. Put $A = A' + 1$, $B = B' + 1$ and $C = C' + 1$. The triples we want then correspond to non-negative solutions of $A' + B' + C' = n - 3$. By stars and bars there are $\binom{n-1}{2}$ of these, which we can also see directly: we choose two of the $n - 1$ gaps between adjacent gold coins for the lead coins. Hence, for $n \geq 1$,

$$\mathrm{P}(A \geq 1,\ B \geq 1,\ C \geq 1) = \frac{\binom{n-1}{2}}{\binom{n+2}{2}} = \frac{(n-1)(n-2)}{(n+1)(n+2)}.$$

This is 0 for $n = 1$ and $n = 2$, as it must be, since three pirates can't all have gold if there are fewer than three gold coins. (For $n = 0$ the probability is trivially 0; the formula doesn't apply there, because $\binom{n-1}{2}$ counts gaps that don't exist.)

Checks

- For $n = 1$ the arrangements of the coins are GLL, LGL and LLG, all equally likely. The first pirate gets the gold only in GLL, which has probability $\frac{1}{3} = \frac{1}{1+2}$, and the second only in LGL, also $\frac{1}{3}$.
- For $n = 3$ the only good arrangement is GLGLG, one of the $\binom{5}{2} = 10$ arrangements, and the formula gives $\frac{2 \cdot 1}{4 \cdot 5} = \frac{1}{10}$.
- Exact enumeration for $n = 1, \ldots, 14$ agrees with all three formulae. A simulation of 2×10^5 chests with $n = 5$ gave 0.715, 0.714 and 0.286, against $\frac{5}{7} \approx 0.714$, $\frac{5}{7}$ and $\frac{12}{42} = \frac{2}{7} \approx 0.286$.

Common slips

- Treating the second pirate's draws as starting afresh with a full chest. How many gold coins are left for her depends on how the first pirate did.
- Multiplying $\mathrm{P}(A \geq 1)$, $\mathrm{P}(B \geq 1)$ and $\mathrm{P}(C \geq 1)$ in part (iii). The shares are not independent, because they must add up to n.

Beyond STEP

The correspondence between positions of the lead coins and solutions of $A + B + C = n$ is the "stars and bars" bijection. It shows that the gaps between randomly placed markers are exchangeable. The continuous version, with points dropped uniformly on a stick, leads to the theory of uniform spacings and the Dirichlet distribution, and order statistics of this kind come up throughout first-year probability. Blitzstein and Hwang, *Introduction to Probability*, treats this counting and the symmetry arguments that go with it very well.

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